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$$\begin{aligned} & \int_0^1 \frac{xe^x}{(x+1)^2} dx \\ & \begin{cases} u = xe^x \\ dv = \frac{dx}{(x+1)^2} \end{cases} \Rightarrow \begin{cases} du = (x+1)e^x dx \\ v = -\frac{1}{x+1} \end{cases} \\ & \int_0^1 \frac{xe^x}{(x+1)^2} dx = \left[-\frac{xe^x}{x+1} \right]_0^1 - \int_0^1 \left(-\frac{1}{x+1} \right) (x+1)e^x dx \\ & = \left[-\frac{xe^x}{x+1} \right]_0^1 + \int_0^1 e^x dx \\ & = \left(-\frac{e}{2} - 0 \right) + [e^x]_0^1 \\ & = -\frac{e}{2} + (e - 1) \\ & = \frac{e}{2} - 1 \end{aligned}$$

References